Learning Curves

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One readable curve-and-repeat evidence record.

Compute means/ranges,compare patterns and propose a controlled follow-up.

Keep one habit: W1–W3/W5 → selected repeat-count task → project curve record → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Over**fitting**. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

Overfitting

> **My remembered idea, example and correction**

Metrics Match Consequences

> **My remembered idea, example and correction**

Train a Regression Model

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Main height-MAEcm at training sizes 10/20/40:validation repeats[7, 9, 8], [5, 7, 6], [4, 6, 5], means 8/6/5, minmax 7–9/5–7/4–6, width 2 each; training means 1/2/3. ClinicPtraining7 throughout/validationmean 8 throughout; Qtraining1 throughout/validationmean 9, 8, 8; Rmainpattern. Shading is observed min/max of3 supplied runs, not confidence/future interval.

  Picture uses errors[6, 5, 7]and[4, 5, 3]on fixed 10 cases:means 6/4 errors, accuracy 40%/60%; 30 plant[4, 4, 4]keepsmean 4 with observedwidth 0.

- Calculate the mean and **range** for the 20-row condition.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Main height-MAEcm at training sizes 10/20/40:validation repeats[7, 9, 8], [5, 7, 6], [4, 6, 5], means 8/6/5, minmax 7–9/5–7/4–6, width 2 each; training means 1/2/3. ClinicPtraining7 throughout/validationmean 8 throughout; Qtraining1 throughout/validationmean 9, 8, 8; Rmainpattern. Shading is observed min/max of3 supplied runs, not confidence/future interval.

  Picture uses errors[6, 5, 7]and[4, 5, 3]on fixed 10 cases:means 6/4 errors, accuracy 40%/60%; 30 plant[4, 4, 4]keepsmean 4 with observedwidth 0.

- Why can **training** error rise while validation error falls?

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main height-MAEcm at training sizes 10/20/40:validation repeats[7, 9, 8], [5, 7, 6], [4, 6, 5], means 8/6/5, minmax 7–9/5–7/4–6, width 2 each; training means 1/2/3. ClinicPtraining7 throughout/validationmean 8 throughout; Qtraining1 throughout/validationmean 9, 8, 8; Rmainpattern. Shading is observed min/max of3 supplied runs, not confidence/future interval.

  Picture uses errors[6, 5, 7]and[4, 5, 3]on fixed 10 cases:means 6/4 errors, accuracy 40%/60%; 30 plant[4, 4, 4]keepsmean 4 with observedwidth 0.

- Can the 40-row result guarantee a mean MAE of 2.5 at 80 rows?

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main height-MAEcm at training sizes 10/20/40:validation repeats[7, 9, 8], [5, 7, 6], [4, 6, 5], means 8/6/5, minmax 7–9/5–7/4–6, width 2 each; training means 1/2/3. ClinicPtraining7 throughout/validationmean 8 throughout; Qtraining1 throughout/validationmean 9, 8, 8; Rmainpattern. Shading is observed min/max of3 supplied runs, not confidence/future interval.

  Picture uses errors[6, 5, 7]and[4, 5, 3]on fixed 10 cases:means 6/4 errors, accuracy 40%/60%; 30 plant[4, 4, 4]keepsmean 4 with observedwidth 0.

- Why is an observed **range** not a confidence interval?

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- The selected task supplies validation error counts at training sizes 12 and24, with three repeats each. The validation-set size is not supplied, so compare observed count means/ranges without inventing accuracy percentages. A different training size is not the validation denominator.

  Describe the observed mean/spread change and the limited three-repeat evidence; do not claim a guaranteed future interval or a causal quantity-only effect.

- At sizes 12 and 24, validation errors are [6, 8, 7] and [4, 7, 7]. Calculate means/**range**s and a careful comparison.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Plot Plant Classifier train/validation performance at several **training** sizes and document the chosen next action.
- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.

- I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Read both curves for **Learning curve**s and **variability**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for **Learning curve**s and **variability**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- **Training** sizes 20,40,80 yield validation accuracies 62%,**70%**,73%. Is doubling again guaranteed to yield 76%? Describe a defensible next experiment.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- With20,40,80 **training** leaves, fixed-set validation accuracy is**60%**,**70%**,78%.
- Interpret the curve and uncertainty.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- As **training** examples grow from 50 to 100 to 200, validation accuracy is **60%**, 72%, 78%. What does the curve suggest, and what does it not guarantee?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 28 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- **Training** sizes 20,40,80 yield validation accuracies 62%,**70%**,73%. Is doubling again guaranteed to yield 76%? Describe a defensible next experiment.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- At sizes 12 and 24, validation errors are [6, 8, 7] and [4, 7, 7]. Calculate means/**range**s and a careful comparison.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.
- At 10 **training** plants validation errors across3 runs are [6, 5, 7]; at 20 plants errors [4, 5, 3], on the same 10-case validation set.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/28

Learning Curves

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Summarise first size

Mean6 errors; range5-7, corresponding to mean 40% accuracy.



2. Summarise larger size

Mean4 errors; range3-5, corresponding to mean 60% accuracy.



3. State bounded pattern

More training plants improved the average in these supplied runs; variability remains.

CHANGE ONE THING

A30-plant experiment gives errors [4, 4, 4].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.
- Which labelled step matches this detail? More **training** plants improved the average in these supplied runs; **variability** remains.
- 1. State bounded pattern
- 2. Summarise larger size
- 3. Summarise first size
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: State bounded pattern More **training** plants improved the average in these supplied runs; **variability** remains. Read the picture in order.

More **training** plants improved the average in these supplied runs; **variability** remains. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: A30-plant experiment gives errors [4, 4, 4]. Which conclusion is supported by the supplied facts?
- 1. Mean remains4; this additional size does not improve the observed mean over20 plants.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Mean remains4; this additional size does not improve the observed mean over20 plants.

Learning curves are measured patterns, not a guarantee of improvement at every data size.

Repair and keep the record

Review Over**fitting** (ai-4/27). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does100% **training** accuracy prove Large was always best?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Metrics Match Consequences (ai-4/25). Goal: Apply a declared score threshold including equality. Short practice: Which labels determine the false-alarm count at threshold 0.4?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Regression Model (ai-4/22). Goal: Fit a line from a declared **training** candidate set. Short practice: May day3's evaluation label choose the candidate line?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Learning Curves: the case

AI ACADEMY / AI-4/28 · 01 · LOOK AT THE FACTS

Learning Curves: the case

The situation

Read a learning curve with repeated-run spread rather than drawing a certainty arrow.

Supplied evidence

At 10 training plants validation errors across3 runs are [6, 5, 7]; at 20 plants errors [4, 5, 3], on the same 10-case validation set. All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain learning curves using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.

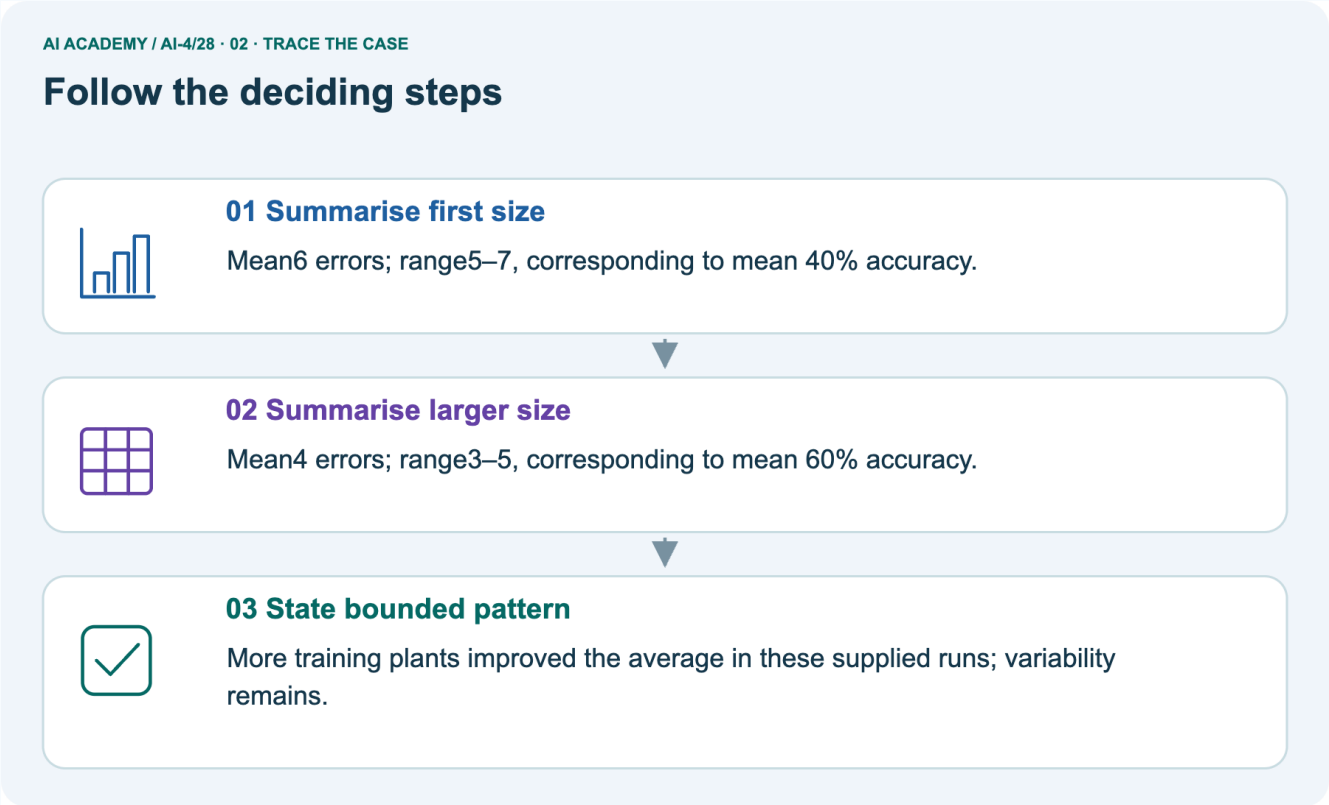
Supplied facts: At 10 **training** plants validation errors across3 runs are [6, 5, 7]; at 20 plants errors [4, 5, 3], on the same 10-case validation set.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Summarise first size: Mean6 errors; **range**5–7, corresponding to mean 40% accuracy. → Summarise larger size: Mean4 errors; **range**3–5, corresponding to mean **60%** accuracy. → State bounded pattern: More **training** plants improved the average in these supplied runs; **variability** remains.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Learning curves and variability at a glance

AI ACADEMY / AI-4/28 · 03 · INSPECT THE LESSON MODEL

Learning curves and variability at a glance

Part	Evidence or action	Interpretation
10 training rows	Validation 7 9 8	Mean 8 range 2
20 training rows	Validation 5 7 6	Mean 6 range 2
40 training rows	Validation 4 6 5	Mean 5 range 2

The decreasing gap is encouraging in this experiment. It does not establish that doubling again will halve error. A range from three repeats is a descriptive spread, not a confidence interval or a guarantee that future results fall inside it. Repeated runs can share records and therefore are not independent studies.

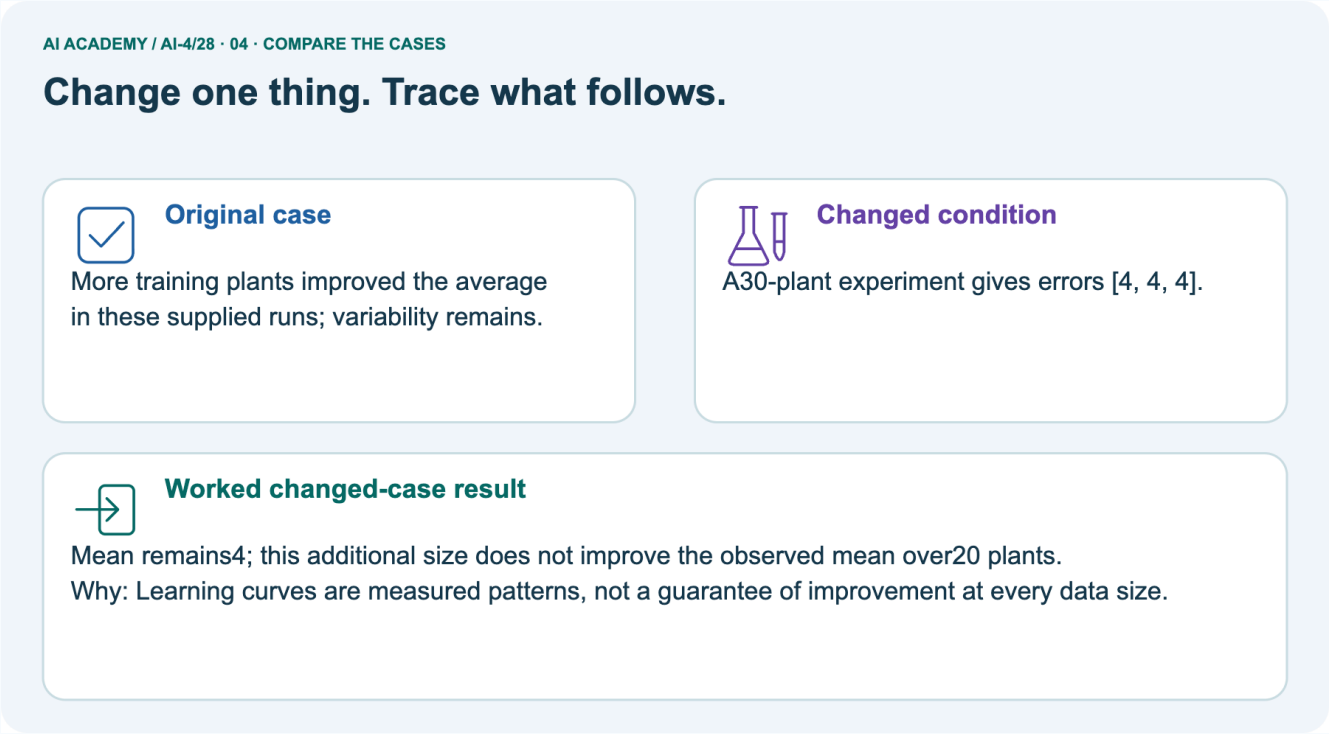
Follow the labelled picture with the matching explanation. Open the vector diagram

Learning curves and **variability** at a glance. 10 **training** rows | Validation 7 9 8 | Mean 8 **range** 2. 20 **training** rows | Validation 5 7 6 | Mean 6 **range** 2. 40 **training** rows | Validation 4 6 5 | Mean 5 **range** 2

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

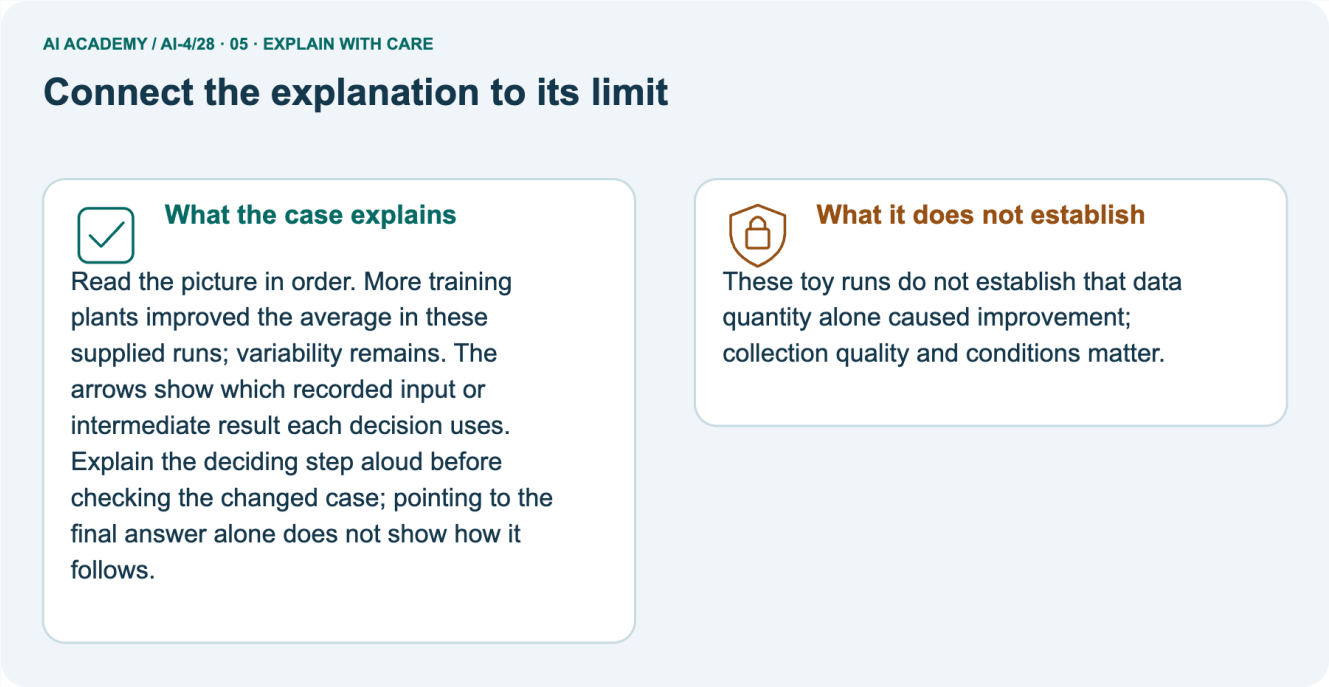
Original case: More **training** plants improved the average in these supplied runs; **variability** remains. Changed condition: A30-plant experiment gives errors [4, 4, 4].

Expected result: Mean remains4; this additional size does not improve the observed mean over20 plants. Reason: **Learning curve**s are measured patterns, not a guarantee of improvement at every data size.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. More **training** plants improved the average in these supplied runs; **variability** remains. The arrows show which recorded **input** or intermediate result each decision uses.

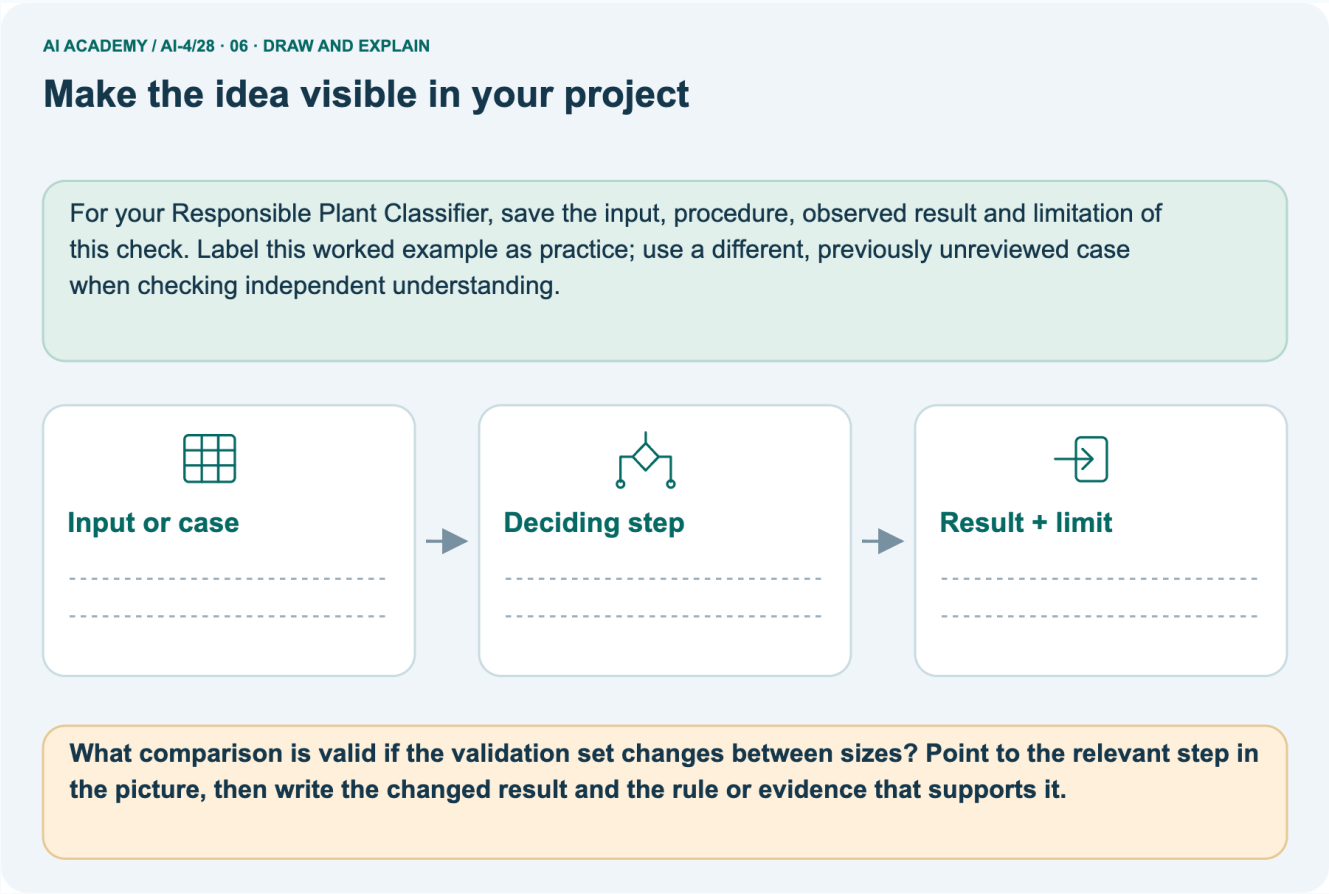
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: These toy runs do not establish that data quantity alone caused improvement; collection quality and conditions matter.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Learning Curves: the case

02. Follow the deciding steps

03. Learning curves and variability at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
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<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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